

Type Checker Disagreement Analysis

Pytifex Research

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1 Code Metrics

File	LOC	Func.	Type Imports	Type Density	Interactions
asyncio-paramspec-partial-callback.py	32	5	5	0.062	1
newtype-typeguard-divergence-refined.py	25	1	4	-0.080	0
paramspec-concatenate-divergence.py	51	7	6	0.098	2
dictionary-comprehension-typeddict-literal-di	45	0	5	0.133	0
asyncio-generic-protocol-callback.py	35	8	6	0.229	3
overload-factory-typeguard-callable.py	57	8	7	0.281	0
overload-classmethod-paramspec.py	56	9	6	0.250	0
self-cyclic-typevar-abstract.py	49	11	5	0.204	0
literal_narrowing.py	16	4	1	0.000	0
walrus.comprehension.py	5	0	0	0.000	0
typeddict_single_field.py	5	1	1	0.000	0

Table 1: Code Metrics for Type Checker Disagreement Examples

2 Type Related Runtime Error Caught (TRREC)

Table 2: TRREC - Checker Error Detection

File	Runtime	Mypy	Pyrefly	Ty	Zuban
asyncio-paramspec-partial-callback.py					✓
newtype-typeguard-divergence-refined.py				✓	
paramspec-concatenate-divergence.py			✓	✓	✓
dictionary-comprehension-typeddict-literal-di				✓	✓
asyncio-generic-protocol-callback.py		✓			✓
overload-factory-typeguard-callable.py				✓	
overload-classmethod-paramspec.py		✓	✓		
self-cyclic-typevar-abstract.py		✓	✓		
literal_narrowing.py		✓		✓	
walrus.comprehension.py			✓	✓	
typeddict_single_field.py		✓	✓	✓	✓

Summary: 0 files cause runtime errors. Mypy detected errors in 5 files, Pyrefly in 5, Ty in 7, Zuban in 5.

3 Disagreement Analysis

Table 3: Disagreement Analysis - Minority Checkers and Verdicts

File	Minority Checkers	Verdict
asyncio-paramspec-partial-callback.py	ZUBAN	ZUBAN: Uncertain
newtype-typeguard-divergence-refined.py	TY	TY: Uncertain
paramspec-concatenate-divergence.py	MYPY	MYPY: Uncertain
dictionary-comprehension-typeddict-literal-di	MYPY, PYREFLY	MYPY: Uncertain; PYREFLY: Uncertain
asyncio-generic-protocol-callback.py	PYREFLY, TY	PYREFLY: Incorrect; TY: Incorrect
overload-factory-typeguard-callable.py	TY	TY: Incorrect
overload-classmethod-paramspec.py	ZUBAN, TY	ZUBAN: Uncertain; TY: Uncertain
self-cyclic-typevar-abstract.py	ZUBAN, TY	ZUBAN: Incorrect; TY: Incorrect
literal_narrowing.py	PYREFLY, ZUBAN	PYREFLY: Correct; ZUBAN: Correct
walrus_comprehension.py	MYPY, ZUBAN	MYPY: Correct; ZUBAN: Correct

Summary: 10 out of 11 files (91%) show disagreement among type checkers.

4 Minority vs Majority Correctness

Table 4: Minority vs Majority Checker Correctness

File	Minority (C/I/U)	Majority (C/I/U)
<i>Note: C=Correct, I=Incorrect, U=Uncertain</i>		
asyncio-paramspec-partial-callback.py	0/0/1	0/0/3
newtype-typeguard-divergence-refined.py	0/0/1	0/0/3
paramspec-concatenate-divergence.py	0/0/1	0/0/3
dictionary-comprehension-typeddict-literal-di	0/0/2	0/2/0
asyncio-generic-protocol-callback.py	0/2/0	2/0/0
overload-factory-typeguard-callable.py	0/1/0	0/3/0
overload-classmethod-paramspec.py	0/0/2	0/0/2
self-cyclic-typevar-abstract.py	0/2/0	0/0/2
literal_narrowing.py	2/0/0	0/2/0
walrus_comprehension.py	2/0/0	0/2/0

Checker	Correct	Incorrect	Uncertain	Total	Accuracy
MYPY	3	2	6	11	27%
PYREFLY	2	3	6	11	18%
TY	0	7	4	11	0%
ZUBAN	4	3	4	11	36%

Table 5: Per-Checker Evaluation Verdicts

5 Per-Checker Evaluation Summary

6 PEP Compliance Analysis

Table 6: PEP Compliance Analysis

File	PEP	Correct?
asyncio-paramspec-partial-callback.py	PEP 484 (ZUBAN)	✓
newtype-typeguard-divergence-refined.py	PEP 484 (TY)	×
paramspec-concatenate-divergence.py	PEP 484 (ZUBAN)	✓
paramspec-concatenate-divergence.py	PEP 484 (TY)	✓
dictionary-comprehension-typeddict-liter	PEP 484 (TY)	✓
dictionary-comprehension-typeddict-liter	PEP 589 (ZUBAN)	✓
dictionary-comprehension-typeddict-liter	PEP 589 (ZUBAN)	✓
dictionary-comprehension-typeddict-liter	PEP 589 (ZUBAN)	✓
asyncio-generic-protocol-callback.py	PEP 484 (ZUBAN)	✓
asyncio-generic-protocol-callback.py	PEP 544 (MYPY)	✓
asyncio-generic-protocol-callback.py	PEP 544 (PYREFLY)	×
asyncio-generic-protocol-callback.py	PEP 544 (ZUBAN)	✓
asyncio-generic-protocol-callback.py	PEP 544 (TY)	×
overload-factory-typeguard-callable.py	PEP 484 (TY)	✓
overload-factory-typeguard-callable.py	PEP 484 (TY)	✓
overload-factory-typeguard-callable.py	PEP 647 (MYPY)	×

File	PEP	Correct?
overload-factory-typeguard-callable.py	PEP 647 (PYREFLY)	×
overload-factory-typeguard-callable.py	PEP 647 (ZUBAN)	×
overload-factory-typeguard-callable.py	PEP 647 (TY)	×
overload-classmethod-paramspec.py	PEP 484 (MYPY)	✓
self-cyclic-typevar-abstract.py	PEP 484 (MYPY)	×
self-cyclic-typevar-abstract.py	PEP 484 (PYREFLY)	×
self-cyclic-typevar-abstract.py	PEP 484 (ZUBAN)	×
self-cyclic-typevar-abstract.py	PEP 484 (TY)	×
literal_narrowing.py	PEP 634 (PYREFLY)	✓
walrus_comprehension.py	PEP 572 (MYPY)	✓
typeddict_single_field.py	PEP 589 (TY)	×